

Importing the Dependencies

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

Data Collection & Pre-Processing

```
# loading the data from csv file to a pandas Dataframe
raw_mail_data = pd.read_csv('/content/mail_data.csv')
```

```
print(raw_mail_data)
```

	Category	Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...
...	...	...
5567	spam	This is the 2nd time we have tried 2 contact u...
5568	ham	Will ü b going to esplanade fr home?
5569	ham	Pity, * was in mood for that. So...any other s...
5570	ham	The guy did some bitching but I acted like i'd...
5571	ham	Rofl. Its true to its name

```
[5572 rows x 2 columns]
```

```
# replace the null values with a null string
mail_data = raw_mail_data.where((pd.notnull(raw_mail_data)), '')
```

```
# printing the first 5 rows of the dataframe
mail_data.head()
```

	Category	Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

```
# checking the number of rows and columns in the dataframe
mail_data.shape
```

```
(5572, 2)
```

Label Encoding

```
# label spam mail as 0; ham mail as 1;

mail_data.loc[mail_data['Category'] == 'spam', 'Category'] = 0
mail_data.loc[mail_data['Category'] == 'ham', 'Category'] = 1
```

spam - 0

ham - 1

```
# separating the data as texts and label

X = mail_data['Message']

Y = mail_data['Category']
```

```
print(X)
```

```
0      Go until jurong point, crazy.. Available only ...
1              Ok lar... Joking wif u oni...
2      Free entry in 2 a wkly comp to win FA Cup fina...
3      U dun say so early hor... U c already then say...
4      Nah I don't think he goes to usf, he lives aro...
...
5567    This is the 2nd time we have tried 2 contact u...
5568              Will ü b going to esplanade fr home?
5569    Pity, * was in mood for that. So...any other s...
5570    The guy did some bitching but I acted like i'd...
5571              Rofl. Its true to its name
Name: Message, Length: 5572, dtype: object
```

```
print(Y)

0      1
1      1
2      0
3      1
4      1
..
5567   0
5568   1
5569   1
5570   1
5571   1
Name: Category, Length: 5572, dtype: object
```

Splitting the data into training data & test data

```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=3)
```

```
print(X.shape)
print(X_train.shape)
print(X_test.shape)
```

```
(5572,)
(4457,)
(1115,)
```

Feature Extraction

```
# transform the text data to feature vectors that can be used as input to the Logistic regression

feature_extraction = TfidfVectorizer(min_df = 1, stop_words='english', lowercase='True')

X_train_features = feature_extraction.fit_transform(X_train)
X_test_features = feature_extraction.transform(X_test)

# convert Y_train and Y_test values as integers

Y_train = Y_train.astype('int')
Y_test = Y_test.astype('int')
```

```
print(X_train)
```

```
print(X_train_features)
```

Training the Model

Logistic Regression

```
model = LogisticRegression()
```

```
# training the Logistic Regression model with the training data
model.fit(X_train_features, Y_train)
```

```
LogisticRegression(C=1.0, class_weight=None, dual=False, fit_intercept=True,
                    intercept_scaling=1, l1_ratio=None, max_iter=100,
                    multi_class='auto', n_jobs=None, penalty='l2',
                    random_state=None, solver='lbfgs', tol=0.0001, verbose=0,
                    warm_start=False)
```

Evaluating the trained model

```
# prediction on training data
```

```
prediction_on_training_data = model.predict(X_train_features)
accuracy_on_training_data = accuracy_score(Y_train, prediction_on_training_data)
```

```
print('Accuracy on training data : ', accuracy_on_training_data)
```

```
Accuracy on training data : 0.9670181736594121
```

```
# prediction on test data
```

```
prediction_on_test_data = model.predict(X_test_features)
accuracy_on_test_data = accuracy_score(Y_test, prediction_on_test_data)
```

```
print('Accuracy on test data : ', accuracy_on_test_data)
```

```
Accuracy on test data : 0.9659192825112107
```

Building a Predictive System

```
input_mail = ["I've been searching for the right words to thank you for this breather. I promise i wont take your help for granted and will fulfil my promise. You have"]

# convert text to feature vectors
input_data_features = feature_extraction.transform(input_mail)

# making prediction

prediction = model.predict(input_data_features)
print(prediction)
```

```
if (prediction[0]==1):  
    print('Ham mail')  
  
else:  
    print('Spam mail')
```

```
[1]  
Ham mail
```

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